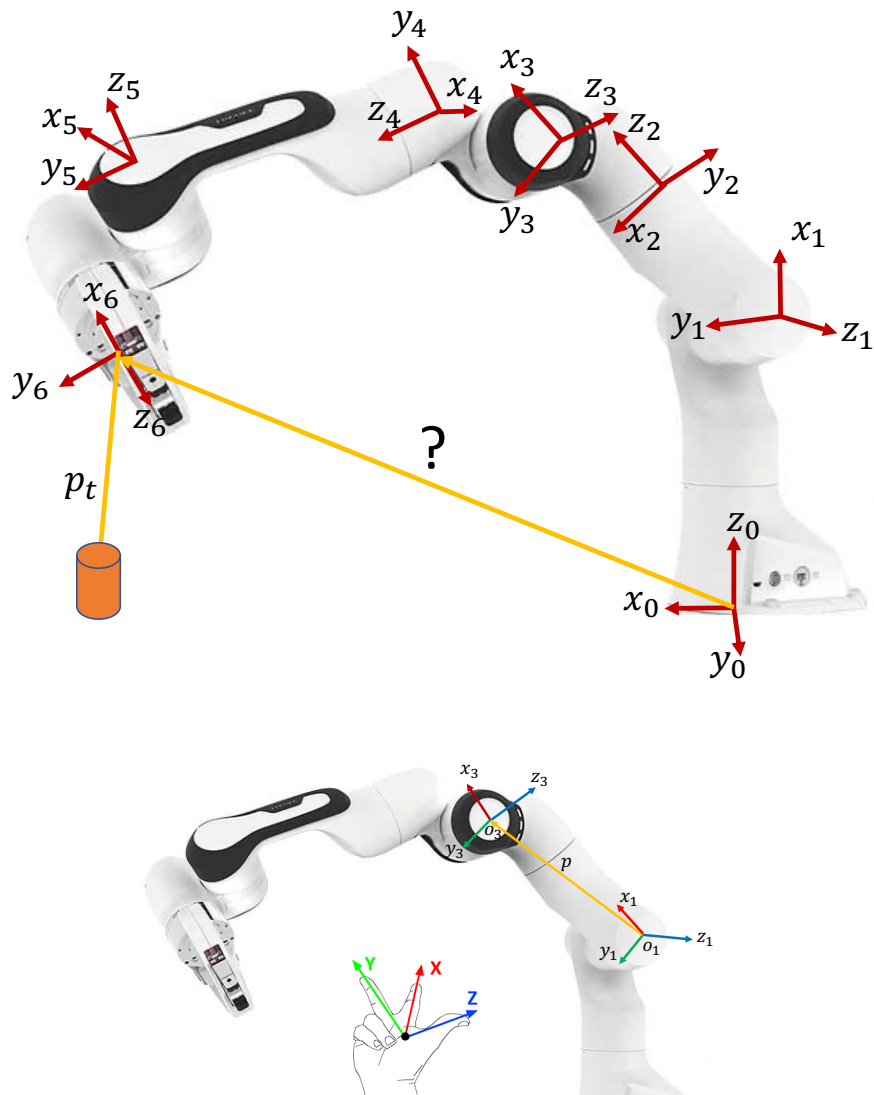


3D Kinematics tutorial-1

Learning outcomes:

- Know what the forward kinematics problem is
- Know how to assign frames to joints in a 3D robot.
- Know what a homogeneous coordinate system is.
- Know how to interpret a homogeneous transformation matrix.

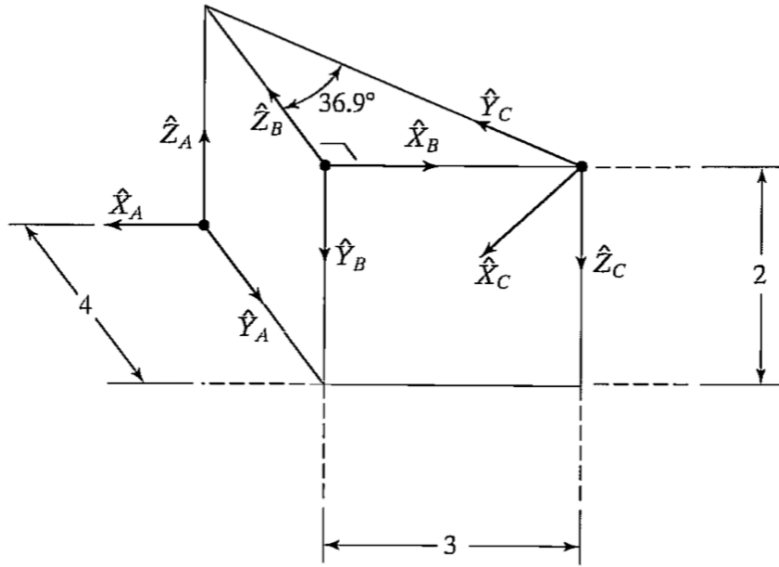
The forward kinematics problem Given knowledge of only the joint angles and link lengths, how to determine the position and orientation of the end-effector of a robot relative to the base.



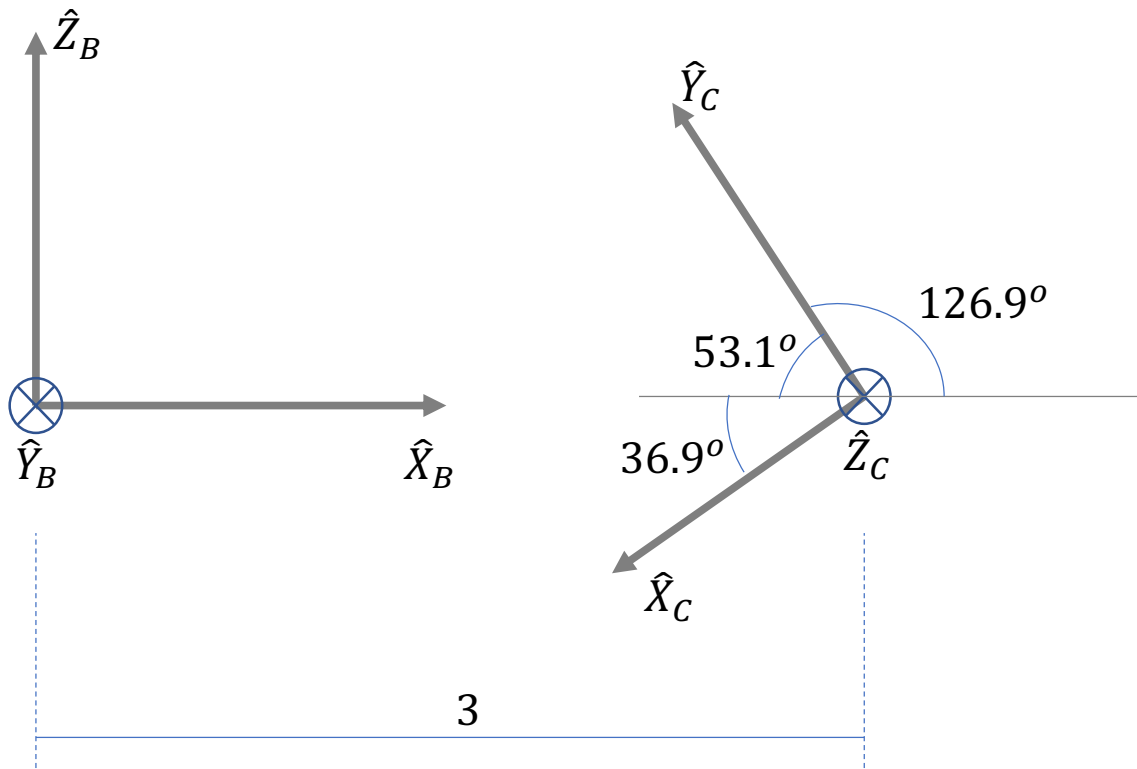
A coordinate frame i consists of an origin, denoted O_i , and three mutually orthogonal basis vectors, denoted (x_i, y_i, z_i) , that are fixed within a given body.

Question-1

Craig, Q 2.32 Find the transformation matrices ${}^B T_C$, ${}^A T_B$, and ${}^A T_C$.



Answer:



$${}^B T_C = \begin{bmatrix} {}^B R_C & {}^B P_C \\ 0_{1 \times 3} & 1 \end{bmatrix}$$

where,

$${}^B R_C = \begin{bmatrix} \hat{X}_C \cdot \hat{X}_B & \hat{Y}_C \cdot \hat{X}_B & \hat{Z}_C \cdot \hat{X}_B \\ \hat{X}_C \cdot \hat{Y}_B & \hat{Y}_C \cdot \hat{Y}_B & \hat{Z}_C \cdot \hat{Y}_B \\ \hat{X}_C \cdot \hat{Z}_B & \hat{Y}_C \cdot \hat{Z}_B & \hat{Z}_C \cdot \hat{Z}_B \end{bmatrix}$$

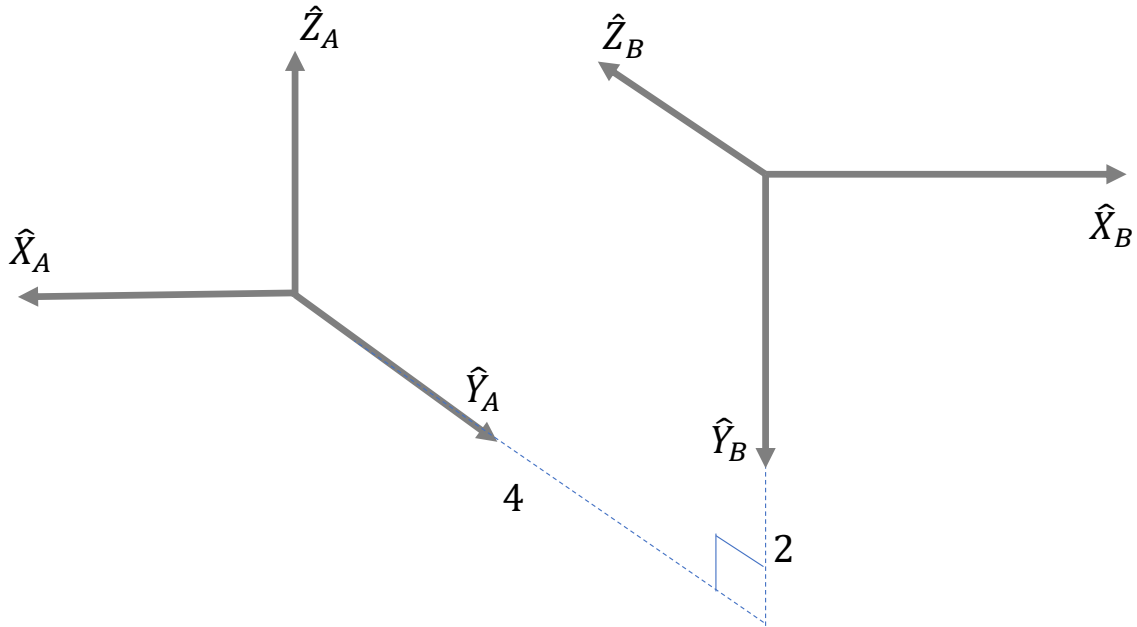
$${}^B R_C = \begin{bmatrix} -0.8 & -0.6 & 0 \\ 0 & 0 & 1 \\ -0.6 & 0.8 & 0 \end{bmatrix}$$

$${}^B P_C = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

Therefore,

$${}^B T_C = \begin{bmatrix} -0.8 & -0.6 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ -0.6 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Similarly,



$${}^A R_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

$${}^A R_B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$${}^A P_B = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$$

$${}^A T_B = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Therefore,

$$\begin{aligned}
 {}^A T_C &= {}^A T_B * {}^B T_C \\
 &= \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.8 & -0.6 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ -0.6 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0.8 & 0.6 & 0 & -3 \\ 0.6 & -0.8 & 0 & 4 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)
 \end{aligned}$$

Question-2

You found a piece of paper with a set of rotation matrices. However, some of the numbers are not clear. Which of the following rotation matrices could be the complete set?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ ? & ? & ? \end{bmatrix} \quad \begin{bmatrix} ? & 0 & 0.7071 \\ ? & 1 & 0 \\ ? & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & ? & 0 \\ 0.5 & ? & 0 \\ 0 & ? & 1 \end{bmatrix}$$

The correct answer is a)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Because a rotation matrix should have a determinant of 1, and inner product of columns or rows should be zero, because it is an orthonormal matrix. Can you guess why a rotation matrix is orthonormal?

Question-3

A 3D rotation matrix is given by $\mathbf{R}_{3 \times 3}$. Given the transformation $\begin{bmatrix} \hat{\mathbf{v}}_{3 \times 1} \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{v}_{3 \times 1} \\ 1 \end{bmatrix}$

what is the value of the ratio given by $\frac{\hat{\mathbf{v}}}{\mathbf{v}}$, where $\mathbf{x}$ is the size of the vector $\mathbf{x}$? Explain your answer.

Answer:

Correct choice is (2).

Reason: The determinant of any rotation matrix is 1. Therefore size of $\mathbf{R}^* \mathbf{v}$ = the size of $\mathbf{v}$.

Question-4

In an $X - Y - Z$ Euler rotation, after doing a rotation around X , the new Y axis is Y' and the new Z axis is Z' . After doing a second rotation around Y' , the new X axis is X'' and the new Z axis is Z'' . There is a vector v such that $v = k_1 X + k_2 Y'$, and k_1, k_2 are real numbers. Is a third rotation around v a valid Euler rotation? Explain your answer.

Choices:

- 1) No
- 2) Yes

Answer:

Correct choice is (1).

Reason: Vector v is linearly dependent on X and Y' . Therefore a 3rd rotation around v is not a valid Euler rotation.

Question-5

Which of the following give the correct expressions for rotation angles given the following rotation matrix?

$$\begin{aligned} {}^i\mathbf{R}_j(\alpha, \beta, \gamma) &= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma - c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} \\ &= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \end{aligned} \quad (2)$$

The correct answer is d)

$$\beta = \text{atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}); \alpha = \text{atan2}\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right); \gamma = \text{atan2}\left(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}\right) \quad (3)$$

Question-6

"A rotation of a frame in 3D space around an axis passing through its origin can be realized through 3 successive rotations around any 3 linearly independent axes". Is this statement correct?

- a) Correct

Question-7

$R^{-1} = R^T$ for any rotation matrix. Therefore, the correct answer: $R^T R^{-1} \neq I$, where I is an identity matrix.

Question-8

Homogeneous coordinates of

$$v = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad (4)$$

is given by

$$v = \begin{bmatrix} ka \\ kb \\ kc \\ k \end{bmatrix} \quad (5)$$

where k is a scalar.

Therefore,

$$v = \begin{bmatrix} 1 \\ 0.5 \\ 1.5 \\ 2 \end{bmatrix} \quad (6)$$

is the correct answer with $k = 2$.

Question-9

Since $T_1 T_2 T_3 = T_4$, $T_3 = (T_1 T_2)^{-1} T_4$.

$$\begin{aligned} T_3 &= (T_1 T_2)^{-1} T_4 \\ &= \begin{bmatrix} R_1 R_2 & (R_1 P_2 + P_1) \\ o & 1 \end{bmatrix}^{-1} \begin{bmatrix} R_4 & P_4 \\ o & 1 \end{bmatrix} \\ &= \begin{bmatrix} (R_1 R_2)^T & -(R_1 R_2)^T (R_1 P_2 + P_1) \\ o & 1 \end{bmatrix} \begin{bmatrix} R_4 & P_4 \\ o & 1 \end{bmatrix} \\ &= \begin{bmatrix} (R_1 R_2)^T R_4 & [(R_1 R_2)^T P_4 - (R_1 R_2)^T (R_1 P_2 + P_1)] \\ o & 1 \end{bmatrix} \end{aligned} \quad (7)$$

This means,

$$\begin{aligned} P_3 &= (R_1 R_2)^T P_4 - (R_1 R_2)^T (R_1 P_2 + P_1) \\ &= -R_2^T R_1^T P_1 - R_2^T R_1^T R_1 P_2 + R_2^T R_1^T P_4 \\ &= -R_2^T R_1^T P_1 - R_2^T P_2 + R_2^T R_1^T P_4 \end{aligned} \quad (8)$$

Therefore c) $P_3 = -R_2^T R_1^T P_1 - R_2^T P_2 + R_2^T R_1^T P_4$ is the correct answer.

Question-10

The correct answer is $q = 0.48i - 0.13j + 0.22k + 0.84$.

$$R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.25 & 0.433 & -0.866 \\ 0.43 & 0.75 & 0.5 \end{bmatrix}.$$

$$\begin{aligned}\epsilon_4 &= \frac{1}{2}\sqrt{1 + R_{11} + R_{22} + R_{33}} \\ &= \frac{1}{2}\sqrt{1 + 0.866 + 0.433 + 0.5} \\ &= 0.8365\end{aligned}\tag{9}$$

$$\begin{aligned}\epsilon_1 &= \frac{R_{32} - R_{23}}{4\epsilon_4} \\ &= \frac{0.75 - (-0.866)}{4 * 0.8365} \\ &= 0.483\end{aligned}\tag{10}$$

$$\begin{aligned}\epsilon_2 &= \frac{R_{13} - R_{31}}{4\epsilon_4} \\ &= \frac{0 - 0.43}{4 * 0.8365} \\ &= -0.1285\end{aligned}\tag{11}$$

$$\begin{aligned}\epsilon_3 &= \frac{R_{21} - R_{12}}{4\epsilon_4} \\ &= \frac{0.25 - (-0.5)}{4 * 0.8365} \\ &= 0.2241\end{aligned}\tag{12}$$

Question-11

If the Jacobian matrix of a 3-link robot manipulator is $J = [J_1 \ J_2 \ J_3]$, where J_i , $i = 1, 2, 3$ are columns of the Jacobian matrix, what is **NOT a condition** for singularity? Explain your answer.

Choices:

- 1) J_1 is colinear with J_3
- 2) $J_2 = 1.5J_3$
- 3) $J_1^T J_2 = 0$
- 4) J_1 is parallel to J_2

Answer:

Correct choice is (3).

Reason: In all other choices, two columns of the Jacobian matrix become parallel or linearly dependent, making the Jacobian matrix singular (two columns behaving like a single column). The condition in the 3rd choice makes two columns perpendicular to each other. This does not make the Jacobian matrix singular.

Question-12

The homogeneous transformation matrix from the end point frame to the base frame of a 3-link manipulator is given by

$${}^0T_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (13)$$

The Jacobian matrix of the above manipulator is given by

a)

$$\mathbf{J} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & -L_2 \sin(\theta_1 + \theta_2) & 0 \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (14)$$

Question-13

What would be the joint torques needed to exert a force vector $\mathbf{f} = [1 \ 1 \ 1]^T$ at the tip of the 3-link manipulator with the homogeneous transformation matrix:

$${}^0T_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (15)$$

b)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) \\ -L_2 \sin(\theta_1 + \theta_2) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (16)$$

Question-14

What condition in joint angles in the 3-link manipulator with the homogeneous transformation matrix given by

$${}^0T_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (17)$$

will result in a torque only around joint-1 to exert a force vector $\mathbf{f} = [1 \ 1 \ 1]^T$ at the tip?

d) $\theta_1 + \theta_2 = \pi/4$

Question-15

Which of the following vectors is in a homogeneous coordinate notation?

b)

$$\mathbf{v}_2 = \begin{bmatrix} 4 & 8 & 2 & 2 \end{bmatrix}^T \quad (18)$$

Question-16

a) What is the purpose of the first rotation of a $Z - Y - X$ Euler rotation?

The objective of the first two rotations is to get one axis to coincide with the corresponding axis of the target frame. Then, since the three axes are orthogonal, one final rotation around the axis that coincides with the target axis will make all three axes coincide.

b) Likewise, what is the purpose of the second rotation?

Question-17

How many sequences are there for Euler rotations? i.e. $Z - Y - X$, $X - Y - Z$, ...

a) For moving frame rotations

$X - Y - Z$

$Y - Z - X$

$Z - X - Y$

$X - Z - Y$

$Z - Y - X$

$Y - X - Z$

in addition;

$X - Y - X$

$X - Z - X$

$Y - X - Y$

$Y - Z - Y$

$Z - X - Z$

$Z - Y - Z$

b) For fixed frame rotations

$$X - Y - Z$$

$$Y - Z - X$$

$$Z - X - Y$$

$$X - Z - Y$$

$$Z - Y - X$$

$$Y - X - Z$$